Bauhaus-Universität Weimar

JOHNS HOPKINS UNIVERSITY

From FEM to Neural Operators: Scientific Machine Learning for Computational Mechanics

From Strong to Weak Form: The Basis of the Finite Element Method

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Motivation

Why weak forms?

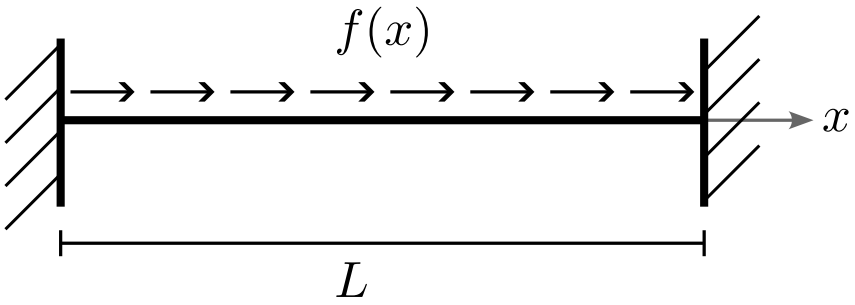
- Physical equations (PDEs) are usually given in **strong form**.
- Strong forms require smooth solutions — often too restrictive.
- The **weak form** relaxes this requirement and is the starting point for FEM approximations.

In this part of the course, we use the Poisson equation as our running case study (for both FEM and later PINNs / Neural Operators)

The Strong Form - Possion Equation in 1D

Poisson Equation

$$-\frac{d^2u}{dx^2} = f(x), \quad x \in (0, L), \qquad u(0) = 0,\ u(L) = 0$$



Requirements:

- $u(x)$ must be twice differentiable
- The equation holds pointwise

From Strong to Weak Form

General steps for obtaining the weak form:

1. Multiply by a test function ($v(x)$ with $v(0) = v(L) = 0$) and integrate over $x \in (0, L)$:

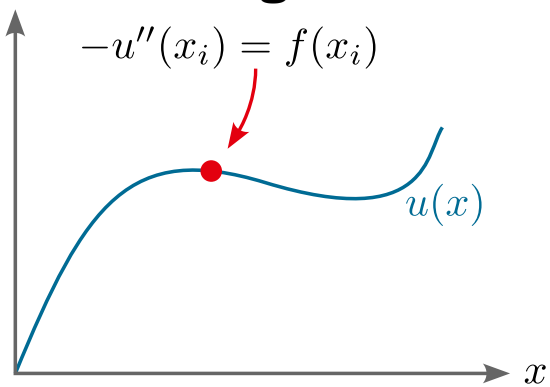
$$\int_0^L (-u''v)\,dx = \int_0^L fv\,dx$$

2. Integrate by parts:

$$\int_0^L u'v'\,dx = \int_0^L fv\,dx$$

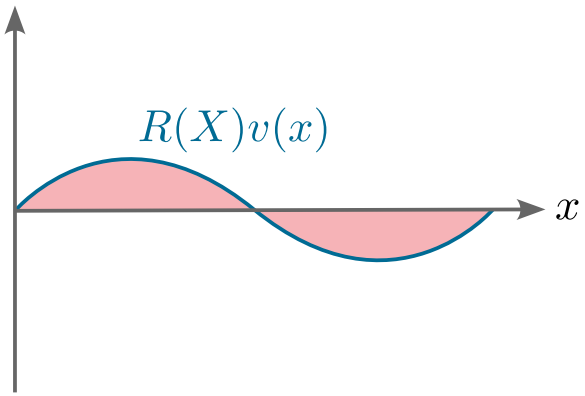
Result: Weak form reduces derivative order on u.

Strong Form



Pointwise Equilibium

Weak Form



Weighted Average = 0

Integration by Parts and Boundary Terms

Integrating $\int_0^L (-u''v)\,dx$ by parts:

$$\int_0^L -u''v\,dx = [-u'v]_0^L + \int_0^L u'v'\,dx$$

For $v(0) = v(L) = 0$ (Pure Dirichlet Problem):

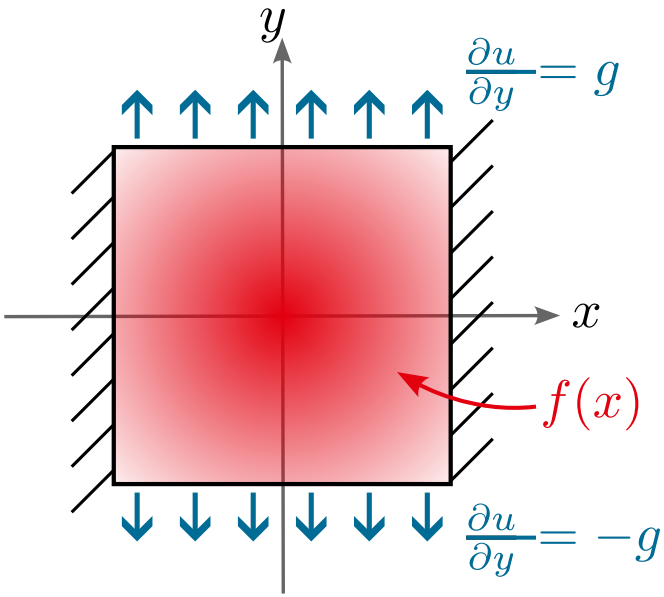
$$\int_0^L u'v'\,dx = \int_0^L fv\,dx$$

For Neumann BC at $x = L$ ($u'(x = L) = g$):

$$\int_0^L u'v'\,dx = \int_0^L fv\,dx + gv(L)$$

- *Dirichlet BCs: imposed on the function space $\Rightarrow$ We only allow test functions that already satisfy $u = 0$ at the Dirichlet boundary.*
- *Neumann BCs: appear naturally.*

Extension to 2D



$$-\Delta u = f \quad \text{in } \Omega$$

$$u = 0 \text{ on } \Gamma_D, \quad \nabla u \cdot \mathbf{n} = g \text{ on } \Gamma_N$$

where $\Gamma_D \Rightarrow$ Dirichlet Boundary

$\Gamma_N \Rightarrow$ Neumann Boundary

Integration by parts in 2D:

$$\int_\Omega -\Delta u v d\Omega = \int_\Omega \nabla u \cdot \nabla v d\Omega - \int_{\partial\Omega} (\nabla u \cdot \mathbf{n}) v d\Gamma$$

If $v = 0$ on Γ_D, and $\nabla u \cdot \mathbf{n} = g$ on Γ_N:

$$\int_\Omega \nabla u \cdot \nabla v \, d\Omega = \int_\Omega f v \, d\Omega + \int_{\Gamma_N} g v \, d\Gamma$$

Example: Constant Load in 1D

$$-u'' = 1, \quad u(0) = u(L) = 0$$

Weak form:

$$\int_0^L u'v' \, dx = \int_0^L v \, dx$$

Exact solution:

$$u(x) = \frac{x(L-x)}{2}$$

The exact solution satisfies both strong and weak forms.

Strong vs Weak: Concept Summary

Strong form

- Equation type $\Rightarrow$ Differential (pointwise)
- Smoothness $\Rightarrow$ Twice differentiable (C^2)
- Boundary Conditions $\Rightarrow$ Dirichlet & Neumann written explicitly in the equation
- FEM compatibility $\Rightarrow \times$

Weak form

- Equation type $\Rightarrow$ Integral (averaged)
- Smoothness $\Rightarrow$ Needs only 1 derivative of u (derivative square-integrable) (H^1)
- Boundary Conditions
  - Dirichlet built into admissible functions (trial/test space)
  - Neumann appears automatically as boundary integrals
- FEM compatibility $\Rightarrow \checkmark$

Closing Remarks

Why strong vs weak forms matter beyond FEM

- FEM: discretize the weak form → element matrices via integrals.
- PINNs: minimize a loss function, which is set in terms of either the strong or the weak form.
- Neural Operators: often trained on data produced from weak-form discretizations.